

1.1. REALNÁ ČÁST

Chceme dokázat

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

Negativně si rozepišme normy:

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle = \langle u+v, u \rangle + \langle u+v, v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} \end{aligned}$$

A protože $x + \bar{x} = 2\operatorname{Re} x$

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + 2\operatorname{Re}\langle u, v \rangle$$

$$\begin{aligned} \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u-v, u \rangle - \langle u-v, v \rangle = \langle u, u \rangle - \langle u, v \rangle - (\langle v, u \rangle - \langle v, v \rangle) = \\ &= \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \overline{\langle u, v \rangle} \end{aligned}$$

A protože $x - \bar{x} = 2i\operatorname{Im} x$

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\operatorname{Re}\langle u, v \rangle$$

A nyní dosazením do pravé strany dokážeme rovnost vlevo:

$$\begin{aligned} \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) &= \frac{1}{4} (\cancel{\|u\|^2} + \cancel{\|v\|^2} + 2\operatorname{Re}\langle u, v \rangle - \cancel{\|u\|^2} - \cancel{\|v\|^2} + 2\operatorname{Re}\langle u, v \rangle) \\ &= \frac{1}{4} (4\operatorname{Re}\langle u, v \rangle) = \operatorname{Re}\langle u, v \rangle \quad \square \end{aligned}$$

IMAGINÁRNÍ ČÁST

$$\operatorname{Im}\langle u, v \rangle = \frac{1}{4i} (\|u-iv\|^2 - \|u+iv\|^2)$$

Opět si rozepišme normy:

$$\begin{aligned} \|u-iv\|^2 &= \langle u-iv, u-iv \rangle = \langle u-iv, u \rangle + \langle u-iv, -iv \rangle = \langle u, u-iv \rangle + \overline{\langle -iv, u-iv \rangle} = \langle u, u \rangle - i\langle u, v \rangle + \overline{\langle -iv, u \rangle} + \\ &+ \overline{\langle -iv, -iv \rangle} = \|u\|^2 + i\overline{\langle u, v \rangle} - i\langle u, v \rangle + \|v\|^2 \\ &= \|u\|^2 + 2\operatorname{Im}\langle u, v \rangle \end{aligned}$$

$$\begin{aligned} \|u+iv\|^2 &= \langle u+iv, u+iv \rangle = \langle u+iv, u \rangle + \langle u+iv, iv \rangle = \langle u, u+iv \rangle + \overline{\langle iv, u+iv \rangle} = \langle u, u \rangle + i\langle u, v \rangle + \\ &+ \overline{\langle iv, u \rangle} + \overline{\langle iv, iv \rangle} = \|u\|^2 - i\overline{\langle u, v \rangle} + i\langle u, v \rangle + \|v\|^2 \\ &= \|u\|^2 - 2\operatorname{Im}\langle u, v \rangle \end{aligned}$$

Z prave strany rovnosti uvoľníme levou:

$$\frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2) = \frac{1}{4}(\cancel{\|u\|^2} + \cancel{\|v\|^2} + 2\operatorname{Im}\langle u, v \rangle - \cancel{\|u\|^2} - \cancel{\|v\|^2} + 2\operatorname{Im}\langle u, v \rangle) =$$

$$\frac{1}{4}(4\operatorname{Im}\langle u, v \rangle) = \operatorname{Im}\langle u, v \rangle \quad \square$$